



深圳市烨新达实业有限公司

SHENZHEN YEXINDA INDUSTRY CO., LTD

液晶显示模组规格确认书

客户	3905000047 YW-G13264C(B2G-G)
产品型号	YXD-13264C-17F (无红外)
关键字描述	6:00' Clock /132*64 STN 黄绿玻璃,/翠绿背光/排线 65mm/无红外/高温高湿低温屏

出厂确认:

技 术	品 管	批 准

客户确认:

研 发	品 管	确 认

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YXD-13264C-17F

LCM 规格书

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规格书修改记录

版本号	修改项目	修改人	修改日期	样品名称或相关标记
VERSION1.0	Preliminary	LZF	2022-04-26	无红外
VERSION1.1	改背光脚	LZF	2024-03-26	背光脚为 PCB 板
VERSION1.2	背面焊点刷三防	LZF	2025-06-16	涂三防

变更说明：PCB 背面的元器件涂抹三防油工艺，高温胶纸去除，焊接 FPC 的点也同步涂抹。



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LCM 使用注意事项



十分感谢您购买我公司的产品,因为液晶模块属于精密器件,结构复杂、工艺精细、所以在使用前请您首先仔细阅读以下注意事项以免给您造成不必要的损失。您在使用过程中遇到困难时,请拨打我们的技术服务电话 0755-86102612,我们将竭诚为您提供服务和帮助。

一、处理保护膜



在装好的模块成品表面贴有一层保护膜,以防在装配时沾污显示屏表面,在整机装配结束前不得揭去,以免弄脏或划伤显示屏表面。

二、加装衬垫

在模块与前面板之间最好加装一块约 0.1 毫米左右的衬垫。

面板还应保持平整,以免在装配后产生扭曲,并可提高其抗振性能。

三、严防静电

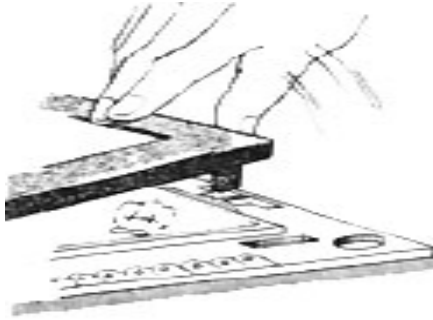


模块中的控制、驱动电路是低压、微功耗的 CMOS 电路,极易被静电击穿,静电击穿是一种不可修复的损坏,而人体有时会产生高达几百伏或几千伏的高压静电,所以,在操作、装配以及使用中都应极其小心,严防静电。为此:

1. 不要用手随意去摸外引线电路板上的电路及金属框
2. 如必须直接接触时应使人体与模块保持在同一电位或将人体良好接地
3. 焊接使用的烙铁和操作用的电动工具必须良好接地没有漏电
4. 不得使用真空吸尘器进行清洁处理因为它会产生很强的静电
5. 空气干燥也会产生静电因此工作间湿度应在 RH60% 左右
6. 取出或放回包装袋或移动位置时也需小心防止产生静电不要随意更换包装或舍弃原包装



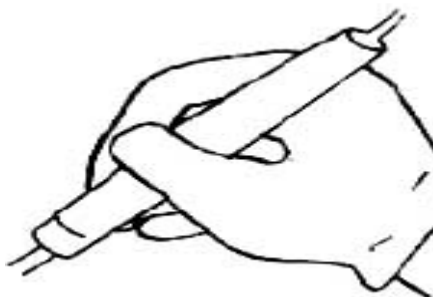
四、装配操作时的注意事项



1. 模块是经过精心设计组装而成的请勿随意自行加工修整
2. 金属框爪不得随意扭动拆卸
3. 不要随意修改加工 PCB 板外形装配孔线路及其部件
4. 不得修改导电胶条
5. 不得修改任何内部支架
6. 不要碰摔折曲扭动模块
7. 安装时不要使 PCB 板受力不均以免造成 PCB 板扭曲使导电连接带拉起

五、焊接

在焊接模块外引线接口电路时应按如下规程进行操作



1. 烙铁头温度小于 280°C
2. 焊接时间小于 $3\sim 4\text{s}$
3. 焊接材料共晶型低熔点
4. 不要使用酸性助焊剂
5. 重复焊接不要超过 3 次且每次重复需间隔 5 分钟

六、模块的使用与保养

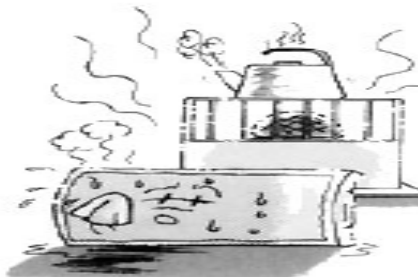
1. 模块的外引线决不允许接错, 在您想调试液晶模块时, 请注意正确接线, 尤其是正、负电源的接线不能有错, 否则可能造成过流、过压、烧毁电路上的芯片等对液晶模块元器件有损的现象;
2. 模块在使用时, 接入电源及断开电源, 必须在正电源稳定接入以后, 才能输入信号电平, 如在电源稳定前或断开后输入信号电平, 有可能损坏模块中的 IC 及电路;
3. 模块使用时, 接入逻辑电源和驱动电源的顺序应是先逻辑电源, 后驱动电源, 断电时, 应先驱动电源, 后逻辑电源。这样做有助于保持屏的良好显示效果和避免在上电、断电时的电压冲击损坏。所以推荐使用带控制的 DC-DC 电源做为模块的驱动电源。
4. 点阵液晶模块显示时的对比度、视角与温度、驱动电压关系很大, 所以, 如果 VEE 调整过高, 不仅会影响显示, 还会缩短模块的使用寿命;
5. 因为液晶材料的物理特性, 液晶的对比度会随着温度的变化而相应变化, 所以, 您加的负电压值应该随温度作相应的调整, 大致是温度变化 10°C 电压变化 1 伏。为满足这一要求, 您可做一个温度补偿电路, 或者安排一个电位器随温度调整负电压值。
6. 不应在规定工作温度范围以外使用, 并且不应在超过存储极限温度的范围外存储, 如果温度低于结晶温度液晶就会结晶, 如果温度过高, 液晶将变成各向同性的液体, 破坏分子取向, 使器件报废;
7. 用力按压显示部分, 会产生异常显示。这时切断电源, 稍待片刻, 重新上电即恢复正常;
8. 液晶显示器件或模块表面结雾时, 不要通电工作, 因为这将引起电极化学反应, 产生断线;



9. 长期用于阳光及强光下时被遮部分会产生残留影像;

七、模块的存储

若长期如几年以上存储我们推荐以下方式



1. 装入聚乙烯口袋最好有防静电涂层并将口封住
2. 在 $-10^{\circ}\text{C} \sim +35^{\circ}\text{C}$ 之间存储
3. 放暗处避强光
4. 决不能在表面压放任何物品
5. 严格避免在极限温/湿度条件下存放

八、责任范围及维修

1. 产品的质量都是通过国际质量认证体系及时间考验的,请用户放心使用
2. 公司出售的产品自出售之日起(保质期为一年)一年之内如出现质量问题给予免费维修。
3. 在您购买液晶显示模块时,焊新达公司将会为您做显示模块的检测,确保您所买的显示模块为完好器件,在您使用过程中因不小心将显示模块损坏(例如静电、焊接连线不当、过流、过压使用等),可送到我们维修部修理(具体事宜请参照深圳市焊新达实业有限公司维修服务规范);如果液晶模块出现屏的问题,比如玻璃面破损、玻璃屏角碎裂等等、将无法进行修理,您的液晶模块只能更换 LCD,我们将只收取 LCD 的成本费用。

1 范围

这个文件包含模块的电特性、机械特性、工作环境包装以及深圳市焊新达实业有限公司生产和质量的常规要求。

2 说明和应用



警告：这种产品上的芯片是静电敏感器件。因此，处理这些部分时，要求非常谨慎。在生产过程中，要确保液晶显示器不要受任何的静电冲击。

3 引用文件

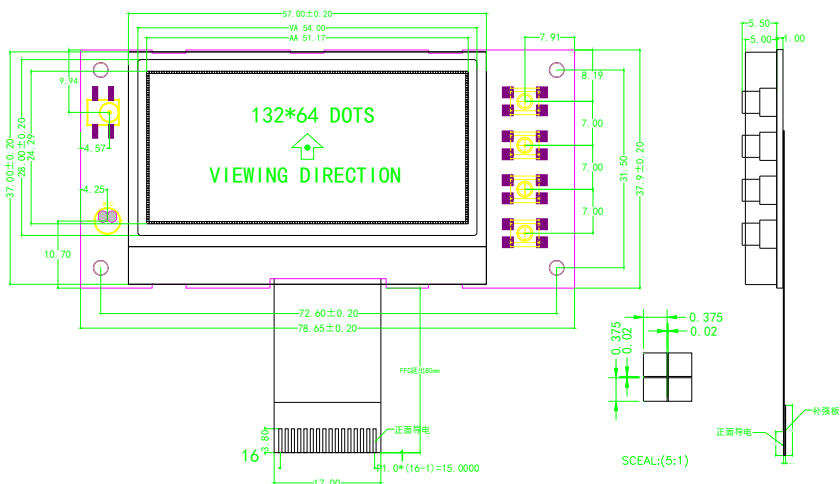
ST7567 规格书 YXD-13264C-17F 设计文件

4 规格特性

4.1 机械规格

类别	标准值	单位
点的数目	132*64	Dots
模块尺寸	78.6 (w)*37.9(h)*7.2(t)	mm
有效显示区	54(w)*28(h)	mm
点大小	0.375(w)*0.375(h)	mm
点间隙	0.395 (w)*0.395(h)	mm
驱动模式	1/64 Duty,1/9 Bias	
显示模式	Positive	
LCM 类型	COG+FPC+PCB	
LCD 类型	STN, Transmissive	
视向	6 :00	
工作温度	-30℃~+70℃	
储藏温度	-40℃~+80℃	
驱动芯片	ST7567 or compatible IC	
背光源	Green(3.3V , max45mA)	

4.2 外形图



4.3 LCD 光电特性

类别	符号	条件	最小值	TYP.	最大值	单位
驱动电压	V	25℃	9.2	9.5	9.7	V
响应时间	Ton	25℃	-	23	400	ms

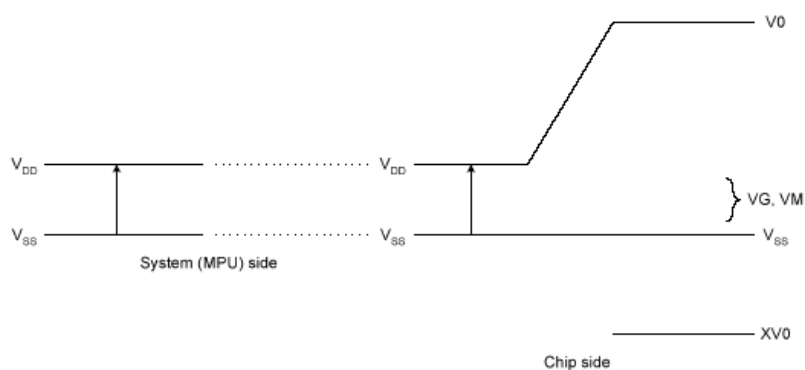


	Toff	25℃	-	239	400	ms
对比度	CR	25℃	3	7	-	-
视角范围		25℃		88		DEG
视角	θ	25℃				DEG
视角	ϕ	25℃				
交叉效应		25℃		1.2		

4.4 绝对最大范围

In accordance with the Absolute Maximum Rating System; please refer to notes 1 and 2.

Parameter	Symbol	Conditions	Unit
Digital Power Supply Voltage	VDD1	-0.3 ~ 3.6	V
Analog Power supply voltage	VDD2, VDD3	-0.3 ~ 3.6	V
LCD Power supply voltage	V0-XV0	-0.3 ~ 16	V
LCD Power supply voltage	VG	-0.3 ~ 3.6	V
LCD Power supply voltage	VM	-0.3 ~ VDD2	V
Input Voltage	Vi	-0.3 ~ VDD1+0.3	V
Operating temperature	TOPR	-30 to +85	°C
Storage temperature	TSTR	-55 to +125	°C



Notes

1. Stresses above those listed under Limiting Values may cause permanent damage to the device.
2. Parameters are valid over operating temperature range unless otherwise specified. All voltages are with respect to VSS unless otherwise noted.
3. Insure the voltage levels of V0, VDD2, VG, VM, VSS and XV0 always match the correct relation:
 $V0 \geq VDD2 > VG > VM > VSS \geq XV0$

4.5 电特性(直流特性)



VSS=0V; Tamb = -30°C to +85°C; unless otherwise specified.

Item	Symbol	Condition		Rating			Unit	Applicable Pin
				Min.	Typ.	Max.		
Operating Voltage (1)	VDD1			1.7	—	3.3	V	VDD1
Operating Voltage (2)	VDD2			2.4	—	3.3	V	VDD2
Operating Voltage (3)	VDD3			2.4	—	3.3	V	VDD3
Input High-level Voltage	V _{IHC}			0.7 x VDD1	—	VDD1	V	MPU Interface
Input Low-level Voltage	V _{ILC}			VSS1	—	0.3 x VDD1	V	MPU Interface
Output High-level Voltage	V _{OHC}	I _{OUT} =1mA, VDD1=1.8V		0.8 x VDD1	—	VDD1	V	D[7:0]
Output Low-level Voltage	V _{OLC}	I _{OUT} =-1mA, VDD1=1.8V		VSS1	—	0.2 x VDD1	V	D[7:0]
Input Leakage Current	I _{LI}			-1.0	—	1.0	μA	MPU Interface
Output Leakage Current	I _{LO}			-3.0	—	3.0	μA	MPU Interface
Liquid Crystal Driver ON Resistance	R _{ON}	Ta=25°C	Vop=8.5V, ΔV=0.85V	—	0.6	0.8	KΩ	COMx
			VG=1.9V, ΔV=0.19V	—	1.3	1.5	KΩ	SEGx
Frame Frequency	FR	Duty=1/65, Vop=8.5V Ta = 25°C		70	75	80	Hz	

Current consumption: During Display, with internal power system, current consumed by whole IC (bare die).

Test Pattern	Symbol	Condition	Rating			Unit	Note
			Min.	Typ.	Max.		
Display Pattern: SNOW (Static)	ISS	VDD1=VDD2=VDD3=3.0V, Booster X5 V _{OP} = 8.5 V, Bias=1/9 Ta=25°C	—	150	300	μA	
Display OFF	ISS	VDD1=VDD2=VDD3=3.0V, Booster X5 V _{OP} = 8.5 V, Bias=1/9 Ta=25°C	—	95	190	uA	
Power Down	ISS	VDD1=VDD2=VDD3=3.0V, Ta=25°C	—	8	16	μA	

Note:

- The Current Consumption is DC characteristics



5 液晶显示器参考驱动条件（交流特性）

AC CHARACTERISTICS

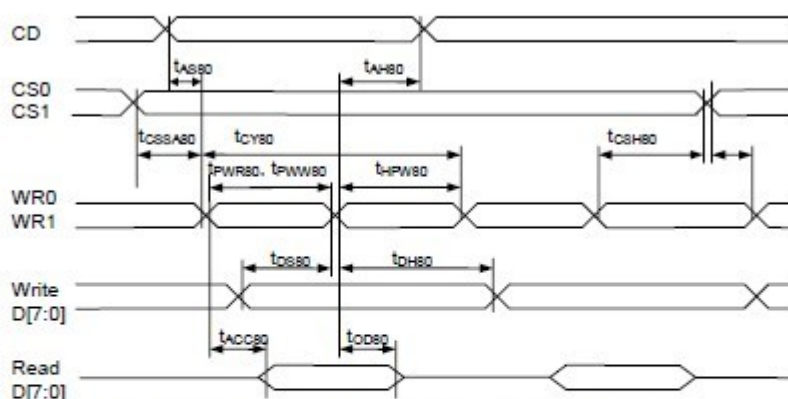


FIGURE 11: Parallel Bus Timing Characteristics (for 8080 MCU)

($2.5V \leq V_{DD} < 3.3V$, $T_a = -30$ to $+85^\circ C$)

Symbol	Signal	Description	Condition	Min.	Max.	Units
t_{AS80}	CD	Address	setup time	0	—	nS
t_{AH80}	CD	Address	hold time	5	—	nS
t_{CSSA80}	CS1/CS0	Chip select	setup time	5	—	nS
t_{CSH80}	CS1/CS0	Chip select	hold time	5	—	nS
t_{CY80}		Cycle time	read	120	—	nS
			write	80	—	nS
t_{PWR80}	WR1	Pulse width	read	60	—	nS
t_{PWW80}	WR0	Pulse width	write	40	—	nS
t_{HPW80}	WR0, WR1	High pulse width	read	60	—	nS
			write	40	—	nS
t_{OS80}	D7~D0	Data	setup time	30	—	nS
t_{OH80}	D7~D0	Data	hold time	0	—	nS
t_{ACC80}		Read access time	$C_L = 100pF$	—	60	nS
t_{OD80}		Output disable time	$C_L = 100pF$	20	—	nS

($1.65V \leq V_{DD} < 2.5V$, $T_a = -30$ to $+85^\circ C$)

Symbol	Signal	Description	Condition	Min.	Max.	Units
t_{AS80}	CD	Address	setup time	0	—	nS
t_{AH80}	CD	Address	hold time	0	—	nS
t_{CSSA80}	CS1/CS0	Chip select	setup time	5	—	nS
t_{CSH80}	CS1/CS0	Chip select	hold time	5	—	nS
t_{CY80}		System cycle time	read	240	—	nS
			write	180	—	nS
t_{PWR80}	WR1	Pulse width	read	120	—	nS
t_{PWW80}	WR0	Pulse width	write	80	—	nS
t_{HPW80}	WR0, WR1	High pulse width	read	120	—	nS
			write	80	—	nS
t_{OS80}	D7~D0	Data	setup time	60	—	nS
t_{OH80}	D7~D0	Data	hold time	0	—	nS
t_{ACC80}		Read access time	$C_L = 100pF$	—	100	nS
t_{OD80}		Output disable time	$C_L = 100pF$	50	—	nS



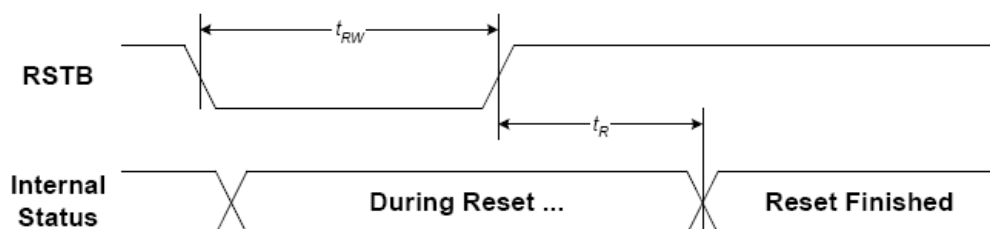
(VDD1 = 1.8V, Ta = 25°C)

Item	Signal	Symbol	Condition	Min.	Max.	Unit
Serial clock period	SCLK	tSCYC		200	—	ns
SCLK "H" pulse width		tSHW		80	—	
SCLK "L" pulse width		tSLW		80	—	
Address setup time	A0	tSAS		60	—	
Address hold time		tSAH		30	—	
Data setup time	SDA	tSDS		60	—	
Data hold time		tSDH		30	—	
CSB-SCLK time	CSB	tCSS		40	—	
CSB-SCLK time		tCSH		100	—	

*1 The input signal rise and fall time (tr, tf) are specified at 15 ns or less.

*2 All timing is specified using 20% and 80% of VDD1 as the standard.

Hardware Reset Timing



(VDD1 = 3.3V, Ta = 25°C)

Item	Symbol	Condition	Min.	Max.	Unit
Reset time	tR		—	1.0	us
Reset "L" pulse width	tRW		1.0	—	

(VDD1 = 2.8V, Ta = 25°C)

Item	Symbol	Condition	Min.	Max.	Unit
Reset time	tR		—	2.0	us
Reset "L" pulse width	tRW		2.0	—	

(VDD1 = 1.8V, Ta = 25°C)

Item	Symbol	Condition	Min.	Max.	Unit
Reset time	tR		—	3.0	us
Reset "L" pulse width	tRW		3.0	—	



Setting RSTB to "L" can initialize internal function. While RSTB is "L", no instruction except read status can be accepted. RSTB pin must connect to the reset pin of MPU and initialization by RSTB pin is essential before operating. Please note the hardware reset is not same as the software reset. When RSTB becomes "L", the hardware reset procedure will start. When RESET instruction is executed, the software reset procedure will start. The procedure is listed below:

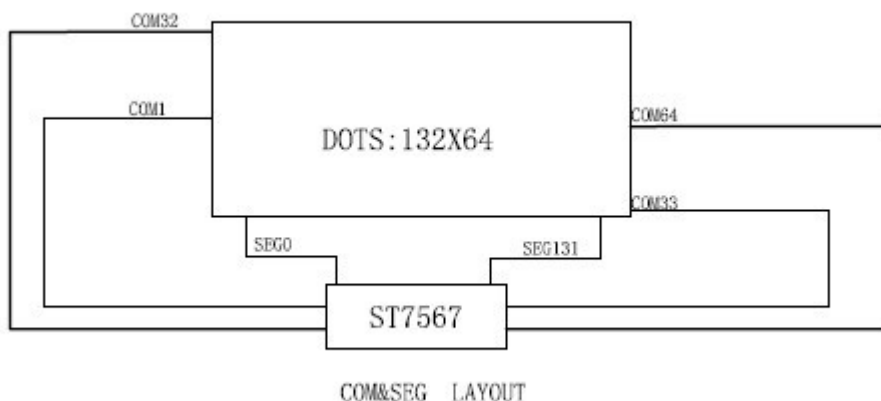
Procedure	Hardware Reset	Software Reset
Display OFF: D=0, all SEGs/COMs output at VSS	V	X
Normal Display: INV=0, AP=0	V	X
SEG Normal Direction: MX=0	V	X
Clear Serial Counter and Shift Register (if using Serial Interface)	V	X
Bias Selection: BS=0	V	X
Booster Level BL=0	V	X
Exit Power Saving Mode	V	X
Power Control OFF: VB=0, VR=0, VF=0	V	X
Exit Read-modify-Write mode	V	V
Start Line S[5:0]=0	V	V
Column Address X[7:0]=0	V	V
Page Address Y[3:0]=0	V	V
COM Normal Direction: MY=0	V	V
V0 Regulation Ratio RR[2:0]=(1,0,0)	V	V
EV[5:0]=(1,0,0,0,0,0)	V	V
Exit Test Mode	V	V

After power-on, RAM data are undefined and the display status is "Display OFF". It's better to initialize whole DDRAM (ex: fill all 00h or write the display pattern) before turning the Display ON. Besides, the power is not stable at the time that the power is just turned ON. A hardware reset is needed to initialize those internal registers after the power is stable.

6. 引脚描述

Pin No.	Symbol	Level	Description
1	VDD	3.3V	Power Supply
2	/RES	H? L, H	LCM Reset Signal Input Terminal
3	A0	H/L	LCM Data/command Signal Input Terminal
4	VDD	3.3V	Power Supply
5	SCK	CLOCK	LCM Serial Clock Input Terminal
6	GND	0V	LCM Power Supply Negative Input Terminal
7	SDA	DATA	LCM Serial Data Input Terminal
8	LEDA	3.3V	LED Backlight Unit Anode Terminal
9	/CS	L	LCM Selection Signal Input Terminal
10	GND	0V	LCM Power Supply Negative Input Terminal
11	VDD	3.3V	Power Supply
12	IR_IN		IR Input
13	KEY_AD		LCM KEY Signal Terminal
14	GND	0V	LCM Power Supply Negative Input Terminal
15	MIC+		Microphone+
16	MIC-		Microphone-

7. 原理框图





8. 命令描述

INSTRUCTION	A0	R/W (RWR)	COMMAND BYTE								DESCRIPTION
			D7	D6	D5	D4	D3	D2	D1	D0	
(1) Display ON/OFF	0	0	1	0	1	0	1	1	1	D	D=1, display ON D=0, display OFF
(2) Set Start Line	0	0	0	1	S5	S4	S3	S2	S1	S0	Set display start line
(3) Set Page Address	0	0	1	0	1	1	Y3	Y2	Y1	Y0	Set page address
(4) Set Column Address	0	0	0	0	0	1	X7	X6	X5	X4	Set column address (MSB)
	0	0	0	0	0	0	X3	X2	X1	X0	Set column address (LSB)
(5) Read Status	0	1	0	MX	D	RST	0	0	0	0	Read IC Status
(6) Write Data	1	0	D7	D6	D5	D4	D3	D2	D1	D0	Write display data to RAM
(7) Read Data	1	1	D7	D6	D5	D4	D3	D2	D1	D0	Read display data from RAM
(8) SEG Direction	0	0	1	0	1	0	0	0	0	MX	Set scan direction of SEG MX=1, reverse direction MX=0, normal direction
(9) Inverse Display	0	0	1	0	1	0	0	1	1	INV	INV =1, inverse display INV =0, normal display
(10) All Pixel ON	0	0	1	0	1	0	0	1	0	AP	AP=1, set all pixel ON AP=0, normal display
(11) Bias Select	0	0	1	0	1	0	0	0	1	BS	Select bias setting 0=1/9; 1=1/7 (at 1/65 duty)
(12) Read-modify-Write	0	0	1	1	1	0	0	0	0	0	Column address increment: Read:+0 , Write:+1
(13) END	0	0	1	1	1	0	1	1	1	0	Exit Read-modify-Write mode
(14) RESET	0	0	1	1	1	0	0	0	1	0	Software reset
(15) COM Direction	0	0	1	1	0	0	MY	-	-	-	Set output direction of COM MY=1, reverse direction MY=0, normal direction
(16) Power Control	0	0	0	0	1	0	1	VB	VR	VF	Control built-in power circuit ON/OFF
(17) Regulation Ratio	0	0	0	0	1	0	0	RR2	RR1	RR0	Select regulation resistor ratio
(18) Set EV	0	0	1	0	0	0	0	0	0	1	Double command!! Set electronic volume (EV) level
	0	0	0	0	EV5	EV4	EV3	EV2	EV1	EV0	
(19) Set Booster	0	0	1	1	1	1	1	1	0	0	Double command!! Set booster level: BL=0: 4X BL=1: 5X
	0	0	0	0	0	0	0	0	0	BL	
(20) Power Save	0	0	Compound Command								Display OFF + All Pixel ON
(21) NOP	0	0	1	1	1	0	0	0	1	1	No operation
(22) Test	0	0	1	1	1	1	1	1	1	-	Do NOT use. Reserved for testing.

Note: Symbol "-" means this bit can be "H" or "L".



Display ON/OFF

The D flag selects the display mode.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	0	1	0	1	1	1	D

D=1: Normal Display Mode.

D=0: Display OFF. All SEGs/COMs output with VSS.

Set Start Line

This instruction sets the line address of the Display Data RAM to determine the initial display line. The display data of specified line address is displayed at the top row (COM0) of the LCD panel.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	1	S5	S4	S3	S2	S1	S0

S5	S4	S3	S2	S1	S0	Line address
0	0	0	0	0	0	0
0	0	0	0	0	1	1
0	0	0	0	1	0	2
0	0	0	0	1	1	3
:	:	:	:	:	:	:
1	1	1	1	0	1	61
1	1	1	1	1	0	62
1	1	1	1	1	1	63

Set Page Address

Y [3:0] defines the Y address vector address of the display RAM.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	0	1	1	Y3	Y2	Y1	Y0

Y3	Y2	Y1	Y0	Page Address	Valid Bit
0	0	0	0	Page0	D0~ D7
0	0	0	1	Page1	D0~ D7
0	0	1	0	Page2	D0~ D7
:	:	:	:	:	:
0	1	1	0	Page6	D0~ D7
0	1	1	1	Page7	D0~ D7
1	0	0	0	Page8 (icon page)	D0



版本号: 1.2

Set Column Address

The range of column address is 0...131. The parameter is separated into 2 instructions. The column address is increased (+1) after each byte of display data access (read/write). This allows MPU accessing DDRAM content continuously. This feature stops at the end of each page (Column Address "83h").

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	0	1	X7	X6	X5	X4

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	0	0	X3	X2	X1	X0

X7	X6	X5	X4	X3	X2	X1	X0	Column address
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	1	0	2
0	0	0	0	0	0	1	1	3
:	:	:	:	:	:	:	:	:
1	0	0	0	0	0	0	1	129
1	0	0	0	0	0	1	0	120
1	0	0	0	0	0	1	1	131

Read Status

Read the internal status of ST7567. The read function is not available in serial interface mode.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	1	0	MX	D	RST	0	0	0	0

Flag	Description
MX	MX=0: Normal direction (SEG0->SEG131) MX=1: Reverse direction (SEG131->SEG0)
D	D=0: Display ON D=1: Display OFF
RST	RST=1: During reset (hardware or software reset) RST=0: Normal operation

Write Data

8-bit data of Display Data from the microprocessor can be written to the RAM location specified by the column address and page address. The column address is increased by 1 automatically so that the microprocessor can continuously write data to the addressed page. During auto-increment, the column address wraps to 0 after the last column is written.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
1	0	Write Data							

Read Data

8-bit data of Display Data from the RAM location specified by the column address and page address can be read to the microprocessor. The read function is not available in serial interface mode.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
1	1	Read Data							



SEG Direction

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	0	1	0	0	0	0	MX

Flag	Description
MX	MX=0: Normal direction (SEG0->SEG131) MX=1: Reverse direction (SEG131->SEG0)

Inverse Display

This instruction changes the selected and non-selected voltage of SEG. The display will be inversed (white -> Black, B -> White) while the display data in the Display Data RAM is never changed.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	0	1	0	0	1	1	INV

Flag	Description
INV	INV=0: Normal display INV=1: Inverse display

All Pixel ON

This instruction will let all segments output the selected voltage and make all pixels turned ON.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	0	1	0	0	1	0	AP

Flag	Description
AP	AP=0: Normal display AP=1: All pixels ON

Bias Select

Select LCD bias ratio of the voltage required for driving the LCD.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	0	1	0	0	0	1	BS

Duty	Bias	
	BS=0	BS=1
1/65	1/9	1/7
1/49	1/8	1/6
1/33	1/6	1/5
1/55	1/8	1/6

Reference LCD Bias Voltage (1/65 Duty with 1/9 Bias)

Symbol	Bias Voltage
V0	V0
VG	$2/9 \times V0$
VM	$1/9 \times V0$
VSS	VSS

Please Note:

* VG range: $1.24V \leq VG < VDD2$.

* VM range: $0.62V \leq VM < VDD2$.

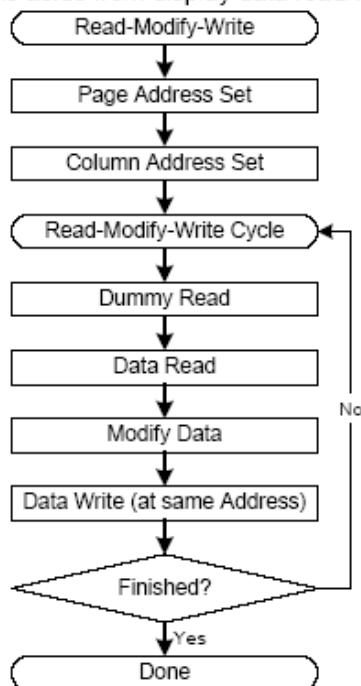


Read-modify-Write

This command is used paired with the "END" instruction. Once this command has been input, the display data read operation will not change the column address, but only the display data write operation will increase the column address ($X[7:0]+1$). This mode is maintained until the END command is input. This function makes it possible to reduce the load on the MPU when there are repeating data changes in a specified display region, such as a blanking cursor.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	1	1	0	0	0	0	0

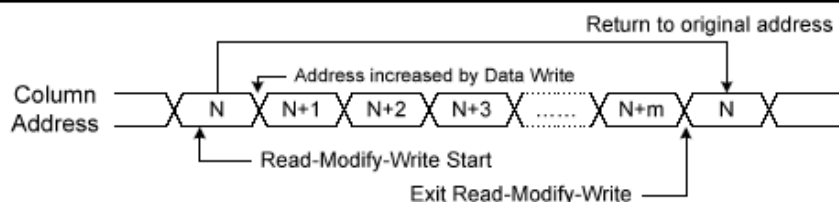
* In Read-modify-Write mode, other instructions aside from display data read/write commands can also be used.



END

When the END command is input, the Read-modify-Write mode is released and the column address returns to the address it was when the Read-modify-Write instruction was entered.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	1	1	0	1	1	1	0



RESET

This instruction resets Start Line ($S[5:0]$), Column Address ($X[7:0]$), Page Address ($Y[3:0]$) and COM Direction (MY) to their default setting. Please note this instruction is not complete same as hardware reset ($RSTB=L$) and cannot initialize the built-in power circuit which is initialized by the RSTB pin. The detailed information is in "Section 7. RESET CIRCUIT".

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	1	1	0	0	0	1	0



COM Direction

This instruction controls the common output status which changes the vertical display direction. The detailed information can be found in Fig 9.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	1	0	0	MY	-	-	-

Flag	Description
MY	MY=0: Normal direction (COM0->COM63) MY=1: Reverse direction (COM63->COM0)

Power Control

This instruction controls the built-in power circuits. Typically, these 3 flags are turned ON at the same time.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	1	0	1	VB	VR	VF

Flag	Description
VB	VB=0: Built-in Booster OFF VB=1: Built-in Booster ON
VR	VR=0: Built-in Regulator OFF VR=1: Built-in Regulator ON
VF	VF=0: Built-in Follower OFF VF=1: Built-in Follower ON

Regulation Ratio

This instruction controls the regulation ratio of the built-in regulator.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	0	0	1	0	0	RR2	RR1	RR0

RR2	RR1	RR0	Regulation Ratio (RR)
0	0	0	3.0
0	0	1	3.5
0	1	0	4.0
0	1	1	4.5
1	0	0	5.0
1	0	1	5.5
1	1	0	6.0
1	1	1	6.5

The operation voltage (V0) calculation formula is shown below: (RR comes from Regulation Ratio, EV comes from EV[5:0])

$V0 = RR \times [1 - (63 - EV) / 162] \times 2.1$, or $V0 = RR \times [(99 + EV) / 162] \times 2.1$

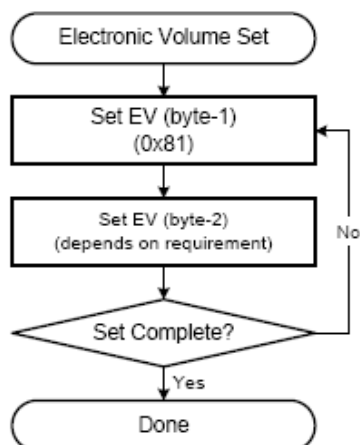
SYMBOL	REGISTER	VALUE
RR	RR[2:0]	3, 3.5, 4, 4.5, 5, 5.5, 6 and 6.5
EV	EV[5:0]	0~63



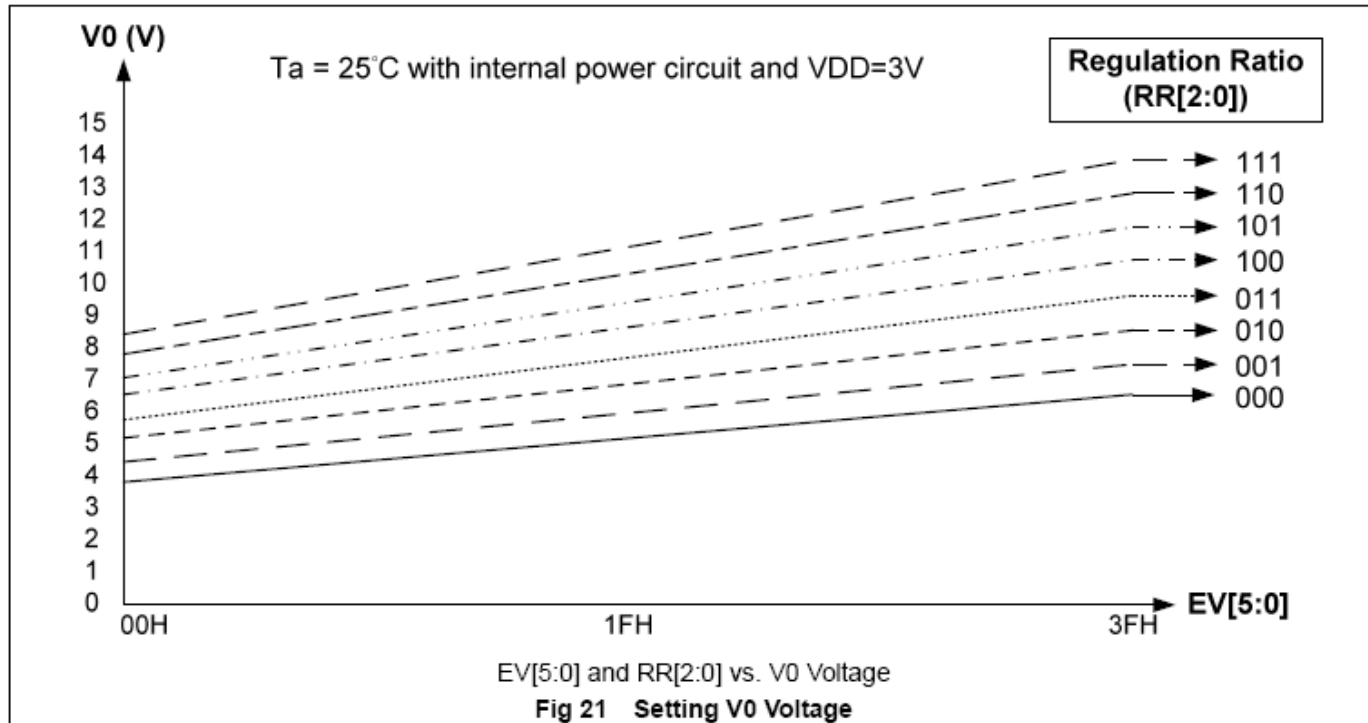
Set EV

This is double byte instruction. The first byte set ST7567 into EV adjust mode and the following instruction will change the EV setting. That means these 2 bytes must be used together. They control the electronic volume to adjust a suitable V0 voltage for the LCD.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	0	0	0	0	0	0	1
0	0	0	0	EV5	EV4	EV3	EV2	EV1	EV0



The maximum voltage that can be generated is dependent on the VDD2 voltage and the loading of LCD module. There are 8 V0 voltage curve can be selected. It is recommended the EV should be close to the center (1FH) for easy contrast adjustment. Please refer to the "Selection of Application Voltage" section for detailed information.

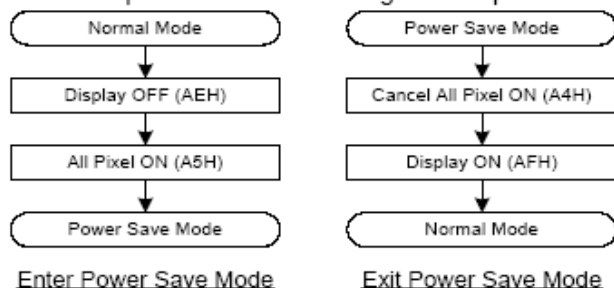




Power Save (Compound Instruction)

This is compound instruction. The 1st instruction is Display OFF (D=0) and the 2nd instruction is All Pixel ON (AP=1). The Power Save mode starts the following procedure: (the display data and register settings are still kept except D-Flag and AP-Flag)

1. Stops internal oscillation circuit;
2. Stops the built-in power circuits;
3. Stops the LCD driving circuits and keeps the common and segment outputs at VSS.

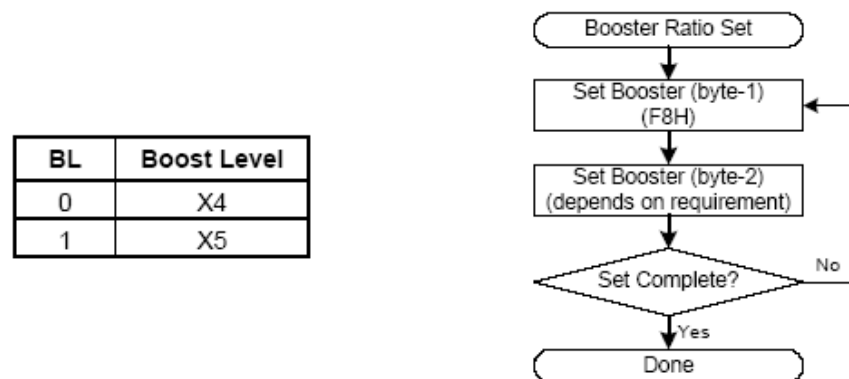


After exiting Power Save mode, the settings will return to be as they were before.

Set Booster

This is double byte instruction. The first byte set ST7567 into booster configuration mode and the following instruction will change the booster setting. That means these 2 bytes must be used together. They control the built-in booster circuit to provide the power source of the built-in regulator. ST7567 booster is built-in booster capacitors. The only external component is a keep capacitor between V0 and XV0. Booster level can be changed with instruction only without changing hardware connection.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	1	1	1	1	0	0	0
0	0	0	0	0	0	0	0	0	BL



BL	Boost Level
0	X4
1	X5

NOP

"No Operation" instruction. ST7567 will do nothing when receiving this instruction.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	1	1	0	0	0	1	1

Test

The test mode is reserved for IC testing. Please don't use this instruction. If the test mode is enabled accidentally, it can be cleared by: issuing an "L" pulse on RSTB pin, issuing RESET instruction or issuing NOP instruction.

A0	R/W(RWR)	D7	D6	D5	D4	D3	D2	D1	D0
0	0	1	1	1	1	1	1	1	-

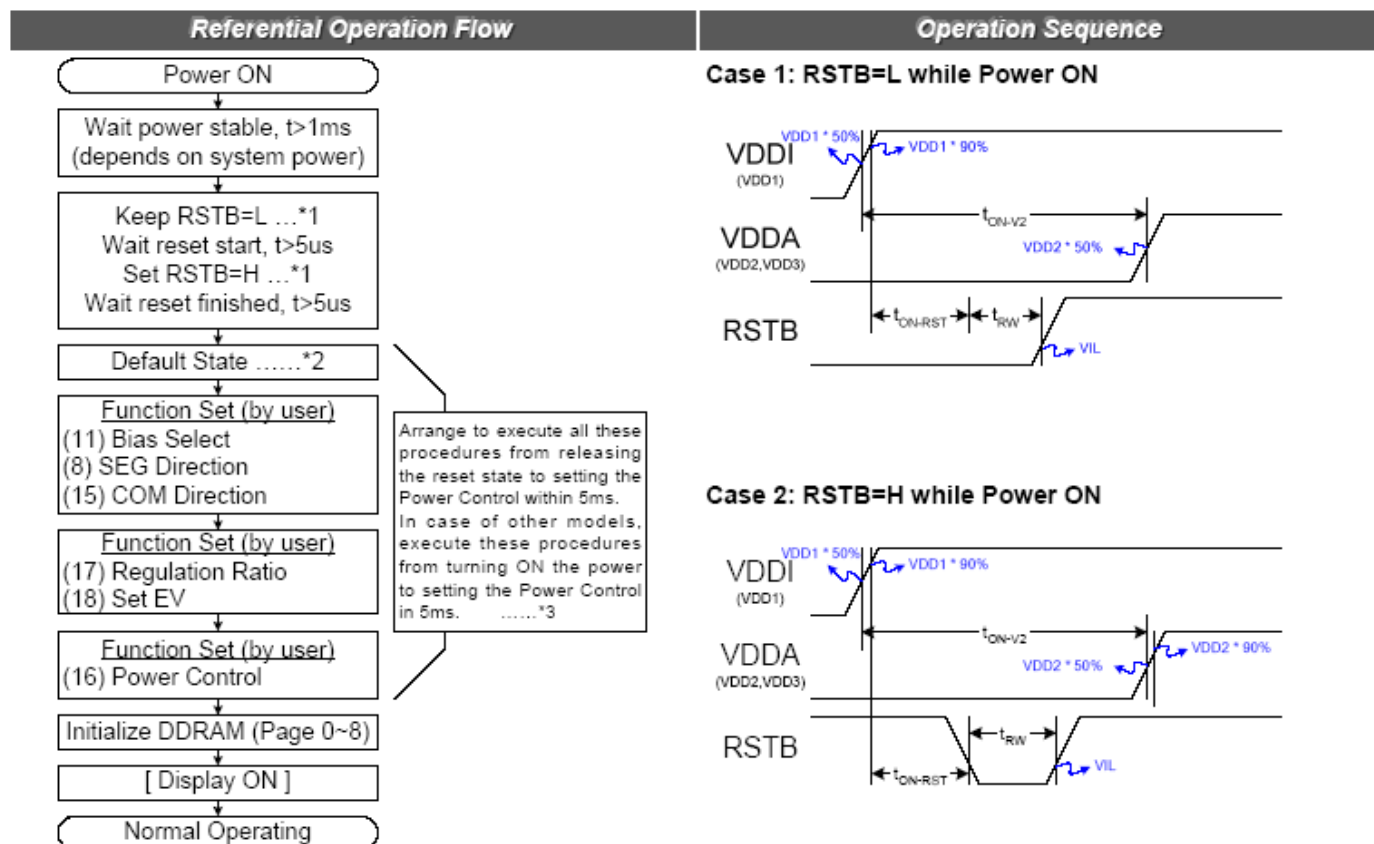
Note: "-" means "1" or "0".



10、操作流程

This section introduces some reference operation flows.

Power ON



Note: The detailed description can be found in the respective sections listed below.

1. Please refer to the timing specification of t_{RW} and t_R .
2. Refer to Section 7. RESET CIRCUIT.
3. The 5ms requirement depends on the characteristics of LCD panel and the external component of the power circuit. It is recommended to check with the real products with external component.
4. The detailed instruction functionality is described in Section 9. INSTRUCTION DESCRIPTION;
5. Power stable is defined as the time that the later power (VDDI or VDDA) reaches 90% of its rated voltage.

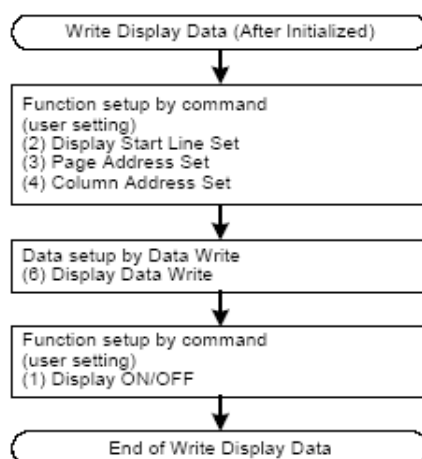
Timing Requirement:

Item	Symbol	Requirement	Note
VDDA power delay	t_{ON-V2}	$0 \leq t_{ON-V2}$	Applying VDDI and VDDA in any order will not damage IC.
RSTB input time	t_{ON-RST}	No Limitation	- If RSTB is Low, High or unstable during power ON, a successful hardware reset by RSTB is required after VDDI is stable. - RSTB=L can be input at any time after power is stable. t_{RW} & t_R should match the timing specification of RSTB. - To prevent abnormal display, the recommended timing is: $0 \leq t_{ON-RST} \leq 30 \text{ ms}$.

- The requirement listed here is to prevent abnormal display on LCD module.



Display Data

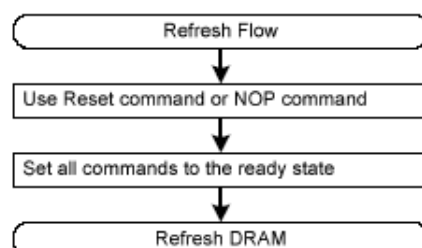


Notes: Reference items

1. The detailed instruction functionality is described in Section 9. INSTRUCTION DESCRIPTION;
2. It is recommended to write display data (initialize DDRAM) before Display ON.

Refresh

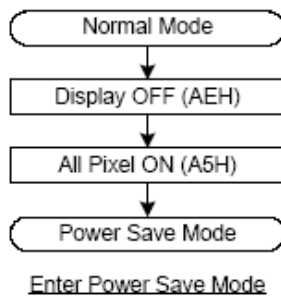
It is recommended to use the refresh sequence regularly in a specified interval.



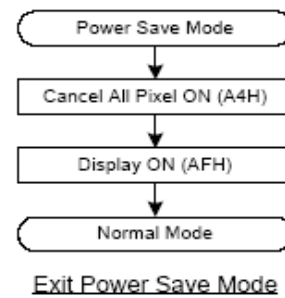


Power-Save Flow and Sequence

ENTERING THE POWER SAVE MODE

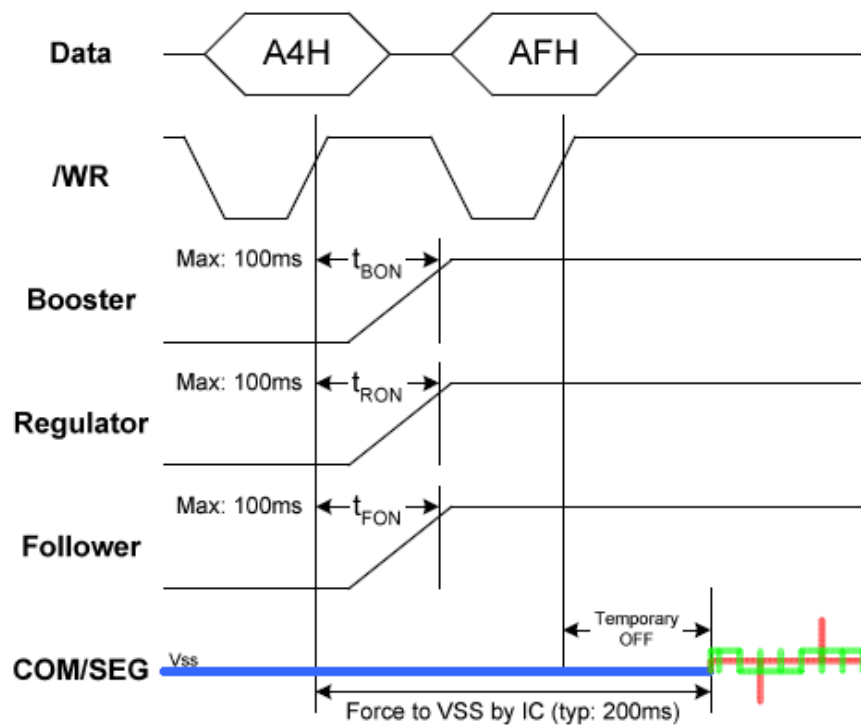


EXITING THE POWER SAVE MODE



INTERNAL SEQUENCE of EXIT POWER SAVE MODE

After receiving "PD=0", the internal circuits (Power) will start the following procedure.



Note:

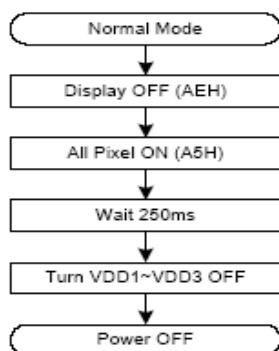
1. The power stable time is determined by LCD panel loading.
2. The power stable time in this figure is base on: LCD Panel Size = 1.4" with C1=1uF, C2=1uF (VDD=2.7V, Vop=9V).



Power OFF Flow and Sequence

In power save mode, LCD outputs are fixed to VSS and all analog outputs are discharged. The power can be turned OFF after ST7567 is in the power save mode. The power save mode can be triggered by the following two methods.

Referential Power OFF Flow



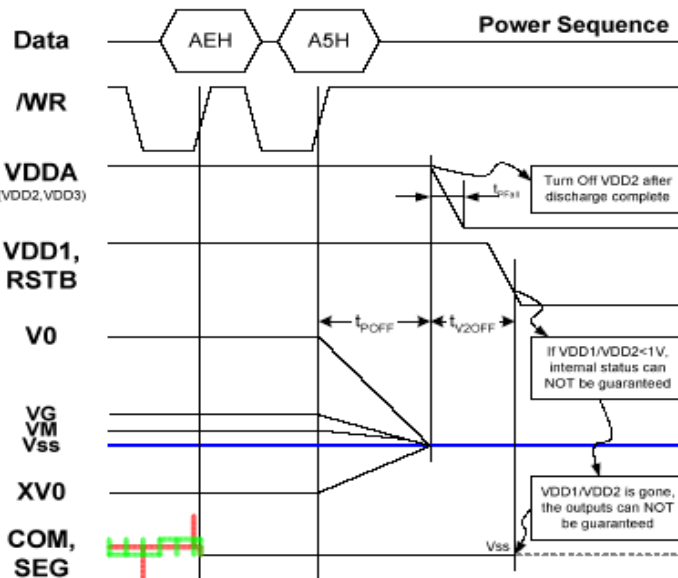
Power OFF Flow

Instruction Flow

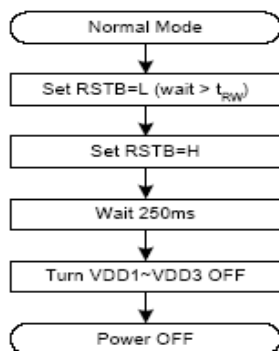
After the built-in power circuits are OFF and completely discharged, the power (VDDI, VDDA) can be removed.

Operation Sequence

CASE 1: Use Power Save Instruction



CASE 2: Use Hardware Reset Function



Power OFF Flow

Instruction Flow

After the built-in power circuits are OFF and completely discharged, the power (VDDI, VDDA) can be removed.

Note:

1. t_{POFF} : Internal Power discharge time. => 250ms (max).
2. t_{V2OFF} : Period between VDDI and VDDA OFF time. => 0 ms (min).
3. It is NOT recommended to turn VDDI OFF before VDDA. Without VDDI, the internal status cannot be guaranteed and internal discharge-process maybe stopped. The un-discharged power maybe flows into COM/SEG output(s) and the liquid crystal in panel maybe polarized.
4. IC will NOT be damaged if either VDDI or VDDA is OFF while another is ON.
5. The timing is dependent on panel loading and the external capacitor(s).
6. The timing in these figures is base on the condition that: LCD Panel Size = 1.4" with C1=1uF, C2=1uF.
7. When turning VDDA OFF, the falling time should follow the specification:
 $20ms \leq t_{Pfall} \leq 0.2sec$

